[33-467] Astrophysics of Stars and the Galaxy

Fall 2023

Instructor:	Katelyn Breivik	**Office:**	Wean 8319
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Class Time:	Tue/Thu 11-12:20	**Class Room:**	DH 2105

Course Description from Course Catalog

The physics of stars is introduced from first principles, leading from star formation to nuclear fusion to late stellar evolution and the end points of stars: white dwarfs, neutron stars and black holes. The theory of stellar structure and evolution is elegant and impressively powerful, bringing together all branches of physics to predict the life cycles of the stars. The basic physical processes in the interstellar medium will also be described, and the role of multi-wavelength astronomy will be used to illustrate our understanding of the structure of the Milky Way Galaxy, from the massive black hole at the center to the halo of dark matter which emcompasses it.

Student Learning Outcomes

At the end of the class, students should be able to:

- ❏ formulate a comphrehensive approach to answering open questions in astronomy that includes using physical prinicples and astronomical data to support their arguments.
- ❏ present research through oral presentations and text format with supporting figures.
- ❏ compute stellar properties from observations and relate the present state of a star to its state at any time in the future.
- ❏ describe how a stellar companion can alter the evolution of stellar populations.
- ❏ understand how stars form and be able to demonstrate how the interstellar medium interacts with the initial and end states of stars.
- ❏ be conversant with the current state of research in extrasolar planets and understand how our solar system is different to and similar to other detected solar systems.
- ❏ understand the structure of the Milky Way galaxy, and able to relate the dark matter distribution to that of stars, gas, dust and black holes.

Textbooks and Course Materials

Canvas: The site can be reached from http://www.cmu.edu/canvas/ The course website will contain the syllabus, homework, project assignments and all required reading. Announcements and visual material presented during lectures will also be posted throughout the semester.

Textbooks: There is no required textbook for this course. All required reading will be available through Canvas. The majority of the stellar structure and evolution material will be drawn from Onno Pols' lecture notes.

Prerequisite Knowledge

The expected prerequisite knowledge for this course is the material covered in 'Stars, Galaxies, and the Universe' (33-224) and its prequisites. Stellar physics builds on a small amount of material from 'Quantum Physics' (33-234) but the required material will also be discussed during this course.

Evaluation Procedures

The course will consist of in-class lectures and homeworks based on the lecture material as well as two long-term projects. There will be no mid-term or final exams. Instead, each project will be evaluated based on two in-class presentations: a mid-project update and a final presentation.

Homework: Homework will be given approximately weekly. The homework will be shorter than for most courses because you will be spending time every week on projects. Science is a collaborative endeavour. To this end, you are encouraged to work with your colleagues on homework and can obtain a maximum of 5 pts extra per homework by citing those who you worked with.

Projects: The majority of the work for this course will be done through two long-term projects to be carried out during the semester. The projects will be chosen from a list, and will be graded based on a mid-project update and a final presentation. The first set of projects is designed to cover material that is complementary to the lectures/readings up to the Fall break. The second list of projects will be relevant for material from after Fall break to the end of the semester.

The goal of these projects is for students to engage with the *process of solving astronomical problems*. To facilitate this, there will be time at the halfway point of each class period ("Project Workshop") to ask questions about projects, brainstorm with fellow students, or carry out research. Many projects will have overlapping tools or concepts so you should plan to work together with your classmates throughout the semester for troubleshooting. You are encouraged to work with a partner (maximum of two people per team!) on your project but this is not a requirement. Note that each project can have a maximum of three teams, and project selection is on a first-come first served bases. You are thus encouraged to not delay selection of the project until the last minute!

Each project has questions that must be answered, but these are not the only questions that *should* be answered. As the project progresses, you will encounter opportunities to dig deeper into some aspects of the problem that you find most interesting. These side quests often lead to the most exciting findings, so you are encouraged to follow them! Throughout the semester, you are also encouraged to chat with me and your classmates about your side quests: this will be facilitated both through questions on the homework as well as playing a role in your mid and final presentations.

Course Component	**Overall Weight**	**Tentative Period**
Individual Homework		
Assignments (8 total)	20%	Week 2-14
Projects		
Project 1: mid-project update	10%	Week 4
Project 1: final presentation	30%	Week 7
Project 2: mid-project update	10%	Week 12
Project 2: final presentation	30%	Week 15

Final grade boundaries are: A: 90 - 100% B: 80 - 89% C: 70 - 79% D: 60 - 69% R: <60%
Boundary cases may be discussed based on demonstrated engagment, effort and improvement over the semester.

Recommended Software and Tools

All of the projects will have a computational component – this is a central facet of astronomical research. Even a pen and paper study must show *some* figures to illustrate your arguments! While there is not a preferred operating system for any of the projects, it is strongly recommended that students use a version control system like `git`, commonly interfaced with web-based services like Github, Gitlab or Bitbucket.

Students are also encouraged to engage with open source astronomical software tools which will aid in their projects. Support for the installation and use of these tools is usually well documented, but students should expect to do some troubleshooting either with the instructor or their peers throughout the semester.

Attendance and Late Work Policies

Formal attendance will not be recorded for the course. However, the material presented in the in-class lectures will be integral to the homework assignments. Class time will also serve as a venue to get help from the instructor and your peers, so in-class participation is likely to correlate strongly with final grade outcomes.

Late homeworks may be submitted for 20% reduced credit up to one week after the due date. After this period, homework will not be accepted. The in-class project times, both at the mid and final points, are critical to your grade so you are encouraged to reach out to the instructor in advance if you expect to not be available.

Accommodations for Students with Disabilities

If you have an accommodations letter from the Disability Resources office, you are encouraged to discuss your accommodations and needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may benefit from accommodations but are not yet registered with the Office of Disability Resources, I encourage you to contact them at access@andrew.cmu.edu.

Statement on Diversity, Equity, and Inclusion

Everyone in this course is must be treated with respect. Science is a human endeavour and we are all accountable for ensuring that the environment we create is welcoming and free from discrimination. Each of us is here for a variety of reasons, comes from a variety of backgrounds, and have a diversity of skills and past experiences. It is expected that the students and instructor will contribute to a collaborative learning environment where diverse perspectives and approaches are celebrated.

Unfortunately, incidents may occur, whether intentional or not, that can create an unwelcoming environment for one or more class members. The university encourages anyone who experiences or observes unfair or hostile treatment on the basis of identity to speak out for justice and support, within the moment of the incident or after the incident has passed. Anyone can share these experiences using the following resources:

- Center for Student Diversity and Inclusion: csdi@andrew.cmu.edu, (412) 268-2150
- Report-It online anonymous reporting platform: reportit.net username: tartans password: plaid

All reports will be documented and deliberated to determine if there should be any following actions. Regardless of incident type, the university will use all shared experiences to transform our campus climate to be more equitable and just.

Communication Expectations

The majority of our communication will be done during classtime but I am also reachable by email at the address listed at the top of this page. I will also hold a weekly office hour on Tuesdays at 2pm. I can also meet outside of those hours, but these meetings will need to scheduled by email.

We will also be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, our grader, and myself. Rather than emailing questions directly to me, I encourage you to post your questions on Piazza. If you run into any issues, please reach out to me so we can get them sorted as soon as possible! Find our class page at:

https://piazza.com/cmu/fall2023/33467a/info

Student Wellness

Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful. If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support including from the Timely Care App. Counseling and Psychological Services (CaPS) is alsoå here to help: call 412-268-2922 and visit their website.

Consider reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help. If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night: CaPS: 412-268-2922 Re:solve Crisis Network: 888-796-8226

The CMU Pantry is committed to reducing hunger among students by providing nutritious food at no cost. Accessing these food resources can promote a healthier, balanced university experience.

Planned Course Content

Week	Topics & Homework
1	Standard Observations, Accessing and Processing Data
2	Equilibrium, Timescales, and Stellar Interiors Project #1 Selection: 09/05 **Assignment:**# 1: 09/07
3	Stellar Interiors and Energy Transport **Assignment:**# 2: 09/14
4	Energy Transport and Nuclear Processes Project #1 Mid-way presentations
5	Nuclear Processes and Low-mass Stars **Assignment:**# 3: 09/26
6	Massive Stars and Stellar Remnants **Assignment:**# 4: 10/05
7	**Project #1 Final presentations**
8	Fall Break
9	Binary Evolution Project #2 Selection: 10/26
10	Gravitational Waves and Star Formation **Assignment:**# 5: 11/02
11	Exoplanets **Assignment:**# 6: 11/09
12	Exoplanets and Galactic Dynamics Project #2 Mid-way presentations
13	Galactic Dynamics **Assignment:**# 7: 11/21
14	Chemical Evolution and Local Group **Assignment:**# 8: 11/30
15	**Project #2 Final presentations**

NOTE: The above actual dates may be modified based on how the semester progresses. **Exact dates and instructions will be announced on course webpage**.